

JOYLIN MATHIAS

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Education

St Joseph Engineering College

Bachelor of Engineering in Computer Science

Sep. 2023 – Present

Mangalore, Karnataka

Relevant Coursework

- Data Structures
- DBMS
- Computer Networks
- Machine Learning
- OOP
- Operating Systems
- Artificial Intelligence
- Software Engineering

Internships

ShadowFox

Web Development Intern

Sep 2025

Remote

- Developed two responsive web applications using HTML, CSS, JavaScript, and React.
- Designed interactive and user-friendly interfaces with emphasis on responsiveness and seamless user experience across devices.
- Deployed a web application using GitHub Pages and gained hands-on experience in version control, deployment workflows, and project hosting.
- Enhanced practical knowledge in front-end development, debugging, and component-based architecture.

Projects

Multi-Agent AI Research Assistant | *Python, FastAPI, React.js, LangChain, ChromaDB,*

June 2026 – Present

- Developing a full-stack AI-powered research assistant using multi-agent architecture to perform intelligent document retrieval, summarization, and contextual question answering.
- Built a Retrieval-Augmented Generation (RAG) pipeline using vector embeddings and ChromaDB for semantic search and accurate context-aware response generation.
- Implemented specialized AI agents for research, fact verification, citation generation, and conversational memory to improve response reliability and workflow efficiency.
- Designed scalable backend APIs using FastAPI and an interactive chat-based frontend using React.js for real-time AI-driven knowledge assistance.

SoundSense: Hearing Screening Platform | *ML, Web Dev, Vercel*

Jan 2026 – Apr 2026

- Developed an AI-powered web-based hearing screening platform for early detection of hearing impairments, improving accessibility to affordable preventive healthcare solutions.
- Implemented an audio signal processing pipeline to analyze user responses across multiple sound frequencies and identify potential hearing abnormalities through pattern-based analysis.
- Deployed the platform on Vercel to enable real-time hearing assessment, result visualization, and personalized hearing health insights for end users.

Technical Skills

Languages: Python, Java, C, JavaScript, SQL, HTML/CSS

Developer Tools: VS Code, Git, GitHub, MySQL Workbench, Google Cloud Platform

Technologies/Frameworks: FastAPI, React.js, Tailwind CSS, LangChain, ChromaDB, Machine Learning, RAG, LLMs

Open Source Contributions

- Contributed to open-source projects by resolving issues and delivering code improvements through pull requests in collaborative development environments.
- Applied Git/GitHub workflows involving branching, version control, issue tracking, and pull request-based code integration.

Certifications and Achievements

- Introduction to Cybersecurity
- Generative AI with Large Language Models.
- Earned multiple achievement badges in Google Cloud Computing Foundations (GCCF).